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Milestone 1 Deliverables for Neighbourly

Submitted By

Team Name: BugsBunny

Team Code: Team-015



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

Apartment Association Management System

Project Details

Software Name: Neighbourly - Apartment Association Management System

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Team Details

Team Name: BugsBunny
Team Code: Team-015

Name	Roll Number	Role *
YASH KARSH	22f3002322	Project Architect
Juhi Parekh	23f2005473	Scrum Master
SAFWAN HUMAYUN	24f1001022	Frontend Developer
SANIYA NAAZ	23f2002592	Backend Developer
Saumya Sarkar	21f3001551	Backend Developer/Software Tester

*These are the team's primary roles, and members will collaborate across tasks as the project continues.

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1. Introduction

Neighborly is a web-based apartment association management platform designed to digitize the day-to-day operations of a single residential apartment community. The platform provides a centralized solution for managing communication, maintenance requests, visitor access, facility bookings, and other essential community services. By replacing fragmented manual processes with a unified web application, the system aims to improve operational efficiency, transparency, and convenience for all stakeholders within the apartment community.

1.1 Problem Statement

Current apartment management practices often rely on WhatsApp groups, phone calls, spreadsheets, and paper registers to coordinate daily activities. These fragmented processes result in communication gaps, limited transparency, increased administrative effort, and difficulty in tracking complaints, bookings, and visitor records. As residential communities grow, managing these operations manually becomes increasingly inefficient and error-prone.

1.2 Project Objectives

The primary objectives of Neighbourly are:

- Digitize apartment association operations through a centralized platform.
- Improve communication between residents, administrators, and support staff.
- Simplify complaint reporting and maintenance tracking.
- Streamline visitor management and facility booking.
- Reduce manual administrative work while improving transparency and security.

2. Identification of Users

The stakeholders of the system have been classified into Primary, Secondary, and Tertiary users based on the frequency of interaction with the platform and their involvement in apartment operations.

2.1 Primary Users

2.1.1 Residents/Apartment Owners

Residents are the primary users of the platform. They interact with the system regularly to report issues, receive updates, manage visitors, and access apartment facilities. Their

overall experience largely determines the effectiveness of the platform.

Key features for residents:

- Report maintenance issues
- Book common facilities
- Receive community announcements
- Aim to reduce delays in complaint resolution and fragmented communication

Neighbourly provides residents with a centralized platform to report issues, receive timely updates, and access community services efficiently.

2.1.2 Apartment Administrators

Administrators oversee apartment operations and coordinate communication between residents, maintenance staff, and security personnel. They are responsible for ensuring smooth community management.

Key features for administrators:

- Manage complaints.
- Publish announcements.
- Assign maintenance staff.
- Coordinate apartment operations.
- Get a way to handle multiple requests simultaneously without using manual processes.

Neighbourly simplifies administrative workflows by providing centralized management of complaints, announcements, and apartment operations.

2.2 Secondary Users

2.2.1 Security Guards

Security guards manage visitor access and maintain records of entries and exits. They contribute to the safety and security of the apartment community.

Key features for security personnel:

- Verify visitors
- Maintain visitor logs
- Coordinate with residents
- Minimise manual verification that causes delays and long queues

The platform enables faster visitor verification through pre-approved visitor information and digital records.

2.2.2 Maintenance Staff

Maintenance personnel are responsible for resolving technical and housekeeping issues reported by residents. Their work directly impacts resident satisfaction.

Key features for maintenance staff:

- Attend to assigned complaints
- Update work status
- Coordinate with administrators
- Reduce resolution due to the present lack of visibility in assigned tasks

Neighbourly provides maintenance staff with organized task assignments and status tracking for improved coordination.

2.3 Tertiary Users

2.3.1 Visitors

Visitors interact with the system only during apartment visits but are affected by the efficiency of the visitor management process.

Key features for visitors/guests:

- Verify identity
- Obtain entry approval
- Aim to avoid manual approvals as much as feasible so as to not bother guest visitors

The platform enables quicker verification and smoother visitor entry through digital pre-registration.

2.3.2 Vendors/Househelp/Service Providers

Vendors such as electricians, plumbers, delivery personnel, and technicians along with househelp, visit the apartment to provide services and require secure access management.

Key features for service staff:

- Verify the purpose of the visit

- Record entry and exit timings
- Reduce delays due to manual documentation

Neighbourly maintains digital records of vendor visits while simplifying authorization and verification procedures.

3. User Research, Interviews & Pain Point Analysis

3.1 Research Objective

The objective of the interviews was to understand the existing workflows followed by different stakeholders, identify recurring pain points, and gather user requirements. Interviewees were encouraged to describe their experiences and challenges rather than suggest software features directly.

3.2 Interview Methodology

The research was conducted through:

- Semi-structured interviews
- Observation of existing workflows
- Interview transcripts documented
- Audio/video recordings collected as supporting evidence

3.3 Interview Summaries

3.3.1 Interview 1 - Resident (Shahista Amber, Izhaar Mansion, Kanpur)

Recording of the interview | Transcript and detailed summary

The interview with Mrs. Shahista Amber explored how routine apartment operations are currently managed. Unlike most residents, she offered insights from both a resident's perspective and through her husband, who actively manages the apartment association. She explained that maintenance requests, utility collections, community announcements, and discussions are coordinated primarily through a common WhatsApp group, while several administrative tasks continue to be handled manually.

The participant expressed overall satisfaction with the apartment's management, particularly appreciating the prompt response to maintenance issues and the proactive role of the security guards in ensuring resident safety. However, visitor approvals, payment tracking, and community coordination still rely heavily on informal communication and manual processes, highlighting opportunities for better organization

as the community expands.

3.3.2 Interview 2 - Resident (Anzar Khan, Shiv Angan, Kanpur)

Recording of the interview | Transcript and detailed summary

The interview with Mr. Anzar Khan focused on understanding the operational challenges experienced in a busy apartment community with frequent daily interactions. The participant receives numerous visitors and deliveries each day, making visitor management one of the most noticeable inconveniences in his routine. He explained that complaint reporting, announcements, visitor approvals, and maintenance payments are currently managed through WhatsApp, phone calls, and physical records.

Among all the discussions, the participant particularly emphasized the lack of transparency in complaint handling and recalled missing an important water outage announcement because it was buried within regular WhatsApp conversations. The interview highlighted how fragmented communication and manual coordination can significantly affect residents' daily lives.

3.3.3 Interview 3 - Resident (Ankit Biswas, The Forty Five, Kolkata)

Recording of the interview | Transcript and detailed summary

The interview with Mr. Ankit Biswas examined the processes surrounding maintenance requests, visitor management, and facility bookings within his apartment community. The participant consistently described how routine services depend on administrators maintaining physical registers and coordinating activities manually.

A notable concern raised during the discussion was the absence of visibility throughout the maintenance process. While complaints are eventually resolved, residents remain unaware of their progress until completion. Additionally, the participant pointed out that manually checking facility availability through phone calls often delays planning and creates unnecessary dependency on apartment administrators.

3.3.4 Interview 4 - Resident (Ayan Biswas, The Forty Five, Kolkata)

Recording of the interview | Transcript and detailed summary

The interview with Mr. Ayan Biswas focused primarily on communication and transparency within apartment management. Unlike previous participants, he emphasized challenges related to community announcements, resident participation, and maintenance expenditure rather than complaint handling.

The participant explained that important notices frequently disappear within active WhatsApp conversations, while previous poll results and maintenance spending records

are not readily accessible to residents. He also described delays caused by manual facility booking and visitor approvals, indicating that information accessibility is one of the primary challenges experienced by residents.

3.3.5 Interview 5 - Apartment Admin (Pavan Shiva, Prime Lavender, Bangalore)

Recording of the interview | Transcript and detailed summary

The interview with Mr. Pavan Shiva, a member of the apartment association committee, provided the administrative perspective on managing a residential community. Unlike the resident interviews, this discussion focused on the operational responsibilities involved in coordinating maintenance staff, managing resident records, publishing announcements, and handling payments.

Mr. Shiva explained that much of the administrative workload stems from maintaining information across spreadsheets, paper files, and WhatsApp conversations. He highlighted the difficulty of tracking complaints, reconciling maintenance payments, and coordinating bookings while ensuring that residents remain informed. The interview demonstrated how fragmented workflows increase administrative effort and reduce overall operational efficiency.

3.4 Pain Point Analysis

3.4.1 Common pain points identified across 5 interviews

The pain point analysis presented in this section is based on insights gathered from user interviews, supported by observation studies and research. While the interviews provided first-hand accounts of existing workflows and challenges, additional analysis was conducted to identify recurring patterns and operational inefficiencies observed in apartment management. This combined approach ensured that the identified requirements reflect both stakeholder experiences and practical system-level considerations.

Area	Pain Point	Stakeholders Affected
Complaint Management	Complaint reporting and follow-up rely on phone calls or WhatsApp, with no structured tracking mechanism.	Residents, Admins, Maintenance Staff
Communication	Important announcements are shared through WhatsApp groups, causing critical messages to get buried among regular conversations.	Residents, Admins

Area	Pain Point	Stakeholders Affected
Visitor Management	Visitor approvals require manual verification through phone calls, leading to delays and increased workload for security staff.	Residents, Security Guards, Visitors
Facility Booking	Facility reservations are maintained manually, requiring residents to contact administrators for availability and booking confirmation.	Residents, Admins
Record Management	Resident records, visitor logs, bookings, and maintenance records are maintained across spreadsheets, registers, or paper documents.	Admins, Security Guards
Transparency	Residents have limited visibility into complaint progress, maintenance expenditure, and booking status.	Residents
Administrative Workload	Apartment administrators spend significant time coordinating complaints, bookings, payments, and communication manually.	Admins
Community Participation	Voting and community discussions rely on WhatsApp or physical meetings, reducing accessibility and maintaining no historical records.	Residents, Admins

3.4.2 Analysis of Findings

- **Manual and Fragmented Processes:** Almost every participant described workflows that rely on WhatsApp messages, phone calls, spreadsheets, and physical registers. Since information is spread across multiple communication channels, there is no centralized source of truth, making apartment management inefficient and difficult to scale.
- **Lack of Transparency:** Residents consistently reported that they receive little or no visibility into the progress of maintenance requests after reporting them. Similarly, maintenance expenditure, booking status, and payment records are not easily accessible, reducing trust and requiring frequent follow-ups with

administrators.

- **Inefficient Communication:** Apartment announcements and notices are primarily communicated through WhatsApp groups. As these groups are also used for general conversations, important notices are often overlooked or difficult to locate later, resulting in poor dissemination of critical information.
- **Dependency on Manual Coordination:** Many routine activities including complaint assignment, visitor approvals, and facility bookings depend heavily on administrators or security guards. This creates bottlenecks whenever responsible personnel are unavailable and increases the overall administrative workload.
- **Limited Digital Record-Keeping:** Visitor logs, booking schedules, maintenance records, and resident information are maintained manually through paper registers or spreadsheets. This makes historical information difficult to retrieve, analyze, or audit, and increases the possibility of human error.
- **Opportunities for Improvement:** The interviews collectively indicate the need for a centralized apartment management platform that streamlines communication, improves transparency, digitizes record management, and reduces dependence on manual coordination while supporting the daily workflows of all stakeholders.

4. User Stories

The following user stories have been derived from the stakeholder interviews and observational research and written using the SMART (Specific, Measurable, Achievable, Relevant, and Time-bound) guidelines. Each story focuses on addressing an identified user need arising from the observed pain points.

4.1 Primary Users

US-01

As a resident,

I want to view the status of my maintenance request,

So that I can track its progress without repeatedly contacting the apartment office.

US-02

As a resident,

I want to reserve shared facilities by selecting an available time slot,

So that booking conflicts are avoided.

US-03

As a resident,

I want to pre-register my visitors before their arrival,

So that they can enter the apartment without unnecessary waiting.

US-04

As a resident,
I want to receive notifications for important apartment updates,
So that I stay informed without checking multiple communication channels.

US-05

As a resident,
I want to view current and previous community announcements.
So that I do not miss important information.

US-06

As an administrator,
I want to assign maintenance requests to the appropriate staff,
So that resident issues are resolved efficiently.

US-07

As an administrator,
I want to publish announcements from a centralized dashboard,
So that all residents receive consistent information.

4.2 Secondary Users

US-08

As a maintenance staff member,
I want to view my assigned tasks and their priorities,
So that I can complete them efficiently.

US-09

As a security guard,
I want to verify visitor details against approved visitor records,
So that only authorized visitors are allowed entry.

US-10

As a security guard,
I want to record visitor entry and exit times digitally,
So that visitor records remain accurate and searchable.

4.3 Tertiary Users

US-11

As a visitor,
I want my identity to be verified promptly,
So that I can enter the apartment without unnecessary waiting.

US-12

As a visitor,
I want my approved visit details to be available to security personnel,
So that the entry process is completed quickly.

US-13

As a visitor,
I want the resident to be notified when I arrive,
So that they can authorize my entry promptly.

US-14

As a vendor,
I want my service visit to be logged digitally,
So that my access history is maintained accurately.

US-15

As a vendor,
I want my visit to be verified before entering the apartment,
So that I can begin my assigned work without delays.

Link to folder of all user interviews, transcripts, and detailed summaries
Link to Honor Codes

--- End of Milestone 1 Report ---